

IMO 2026 — Problem 5

Solution by Claude Fable 5

2026-07-18

Source: `results/claude-fable-5/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)}$$

for every $x, y \in \mathbb{R}_{>0}$.

Approaches tried

- **Squared reformulation + substitution** $x = f(y)$ (successful): squaring both inequalities gives $2x^2 + 2f(y)^2 \geq (f(x) + y)^2 \geq 4xf(y)$. Plugging $x = f(y)$ makes the two bounds collide and forces the exact identity $f(f(y)) = 2f(y) - y$. Orbits of f are then arithmetic progressions; positivity forces $f(y) \geq y$.
- **Testing candidates:** $f(x) = kx$ works only for $k = 1$; but $f(x) = x + c$ works for every constant $c \geq 0$ (QM-AM-GM applied to the pair $x, y + c$). Numerically verified in `code/verify.py`; random perturbations of $x + c$ violate one of the two inequalities.
- **Orbit rigidity / asymptotic sandwich** (successful): a point a with gap $d = f(a) - a > 0$ generates the orbit $a + nd$ on which $f(t) = t + d$. Using orbit points as y in both squared inequalities and choosing the orbit point nearest to x gives the two-sided bound $d - \frac{d^2}{4x} \leq$

$f(x) - x \leq d + \frac{d^2}{16x-4d}$ for $x \geq a + d$. Hence all positive gaps are equal and all large enough x have gap exactly d .

- **Downward bootstrap** (successful): if $f(t) = t + d$ for all $t \geq T$, then taking $x = y + d$ in both squared inequalities pins $f(y) = y + d$ for all $y \geq T - d$; induction covers all of $\mathbb{R}_{>0}$.

Current best

Answer: the solutions are exactly the functions $f(x) = x + c$ for a constant $c \geq 0$. Complete proof below: sufficiency is QM–AM–GM applied to the numbers x and $y + c$; necessity follows from the forced identity $f(f(y)) = 2f(y) - y$ (equality at $x = f(y)$), which makes orbits arithmetic progressions, whence $f(y) \geq y$, then an asymptotic sandwich shows the gap $g = f - \text{id}$ is a constant d on $[a + d, \infty)$ for any a with $g(a) = d > 0$, and a downward bootstrap propagates $f = \text{id} + d$ to all of $\mathbb{R}_{>0}$; if $g \equiv 0$ then $f = \text{id}$ (the case $c = 0$).

Full proof

Answer. The functions satisfying the condition are exactly

$$f(x) = x + c \quad (x > 0),$$

where $c \geq 0$ is an arbitrary nonnegative real constant.

Throughout, since all quantities appearing under square roots and on both sides are positive, the given condition is equivalent to the pair of squared inequalities

$$(1) \quad 2x^2 + 2f(y)^2 \geq (f(x) + y)^2, \quad (2) \quad (f(x) + y)^2 \geq 4xf(y),$$

required for all $x, y > 0$.

Part I: every $f(x) = x + c$ with $c \geq 0$ is a solution

Fix $c \geq 0$ and let $f(x) = x + c$; note f maps $\mathbb{R}_{>0}$ to $\mathbb{R}_{>0}$. For $x, y > 0$ put $u = x > 0$ and $v = y + c > 0$. Then

$$\frac{x^2 + f(y)^2}{2} = \frac{u^2 + v^2}{2}, \quad \frac{f(x) + y}{2} = \frac{x + c + y}{2} = \frac{u + v}{2}, \quad xf(y) = uv,$$

so the required chain is exactly the QM–AM–GM inequality for the two positive numbers u, v :

$$\sqrt{\frac{u^2 + v^2}{2}} \geq \frac{u + v}{2} \geq \sqrt{uv},$$

which holds because $2(u^2 + v^2) - (u + v)^2 = (u - v)^2 \geq 0$ and $(u + v)^2 - 4uv = (u - v)^2 \geq 0$. ■

Part II: every solution is of this form

Let f satisfy (1) and (2) for all $x, y > 0$.

Step 1: $f(f(y)) = 2f(y) - y$ for all $y > 0$. Substitute $x = f(y)$ (a legitimate positive value) into (1):

$$2f(y)^2 + 2f(y)^2 \geq (f(f(y)) + y)^2 \implies 2f(y) \geq f(f(y)) + y,$$

using that both sides are positive. Substitute $x = f(y)$ into (2):

$$(f(f(y)) + y)^2 \geq 4f(y) \cdot f(y) \implies f(f(y)) + y \geq 2f(y).$$

Combining, $f(f(y)) + y = 2f(y)$, i.e.

$$(3) \quad f(f(y)) = 2f(y) - y \quad \text{for all } y > 0.$$

Step 2: $f(y) \geq y$; **orbits are arithmetic progressions.** Fix $y > 0$ and define $y_0 = y$, $y_{n+1} = f(y_n)$ ($n \geq 0$); all $y_n > 0$. Applying (3) at the point y_n gives $y_{n+2} = 2y_{n+1} - y_n$, so (y_n) is an arithmetic progression:

$$y_n = y + n g(y), \quad \text{where } g(y) := f(y) - y.$$

If $g(y) < 0$ then $y_n < 0$ for large n , contradicting $y_n > 0$. Hence

$$(4) \quad f(y) \geq y \quad \text{for all } y > 0,$$

and moreover, since $y_{n+1} = f(y_n)$,

$$(5) \quad f(y + n g(y)) = y + (n + 1) g(y) \quad (n = 0, 1, 2, \dots).$$

Two elementary square-root estimates. (E1) For all real u with $-1 \leq u \leq 3$: $\sqrt{1+u} \geq 1 + \frac{u}{2} - \frac{u^2}{2}$.

Proof. If the right side is negative, this is trivial. Otherwise both sides are nonnegative and squaring is equivalent; expanding,

$$(1+u) - \left(1 + \frac{u}{2} - \frac{u^2}{2}\right)^2 = \frac{3}{4}u^2 + \frac{1}{2}u^3 - \frac{1}{4}u^4 = -\frac{u^2}{4}(u-3)(u+1) \geq 0$$

for $-1 \leq u \leq 3$. $\square$

(E2) For $A > 0$, $B \geq 0$: $\sqrt{A^2 + B} \leq A + \frac{B}{2A}$.

Proof. $(A + \frac{B}{2A})^2 = A^2 + B + \frac{B^2}{4A^2} \geq A^2 + B$, and both sides are positive. $\square$

Step 3: a point with gap $d > 0$ pins down all gaps at large arguments. Suppose $g(a) = d > 0$ for some $a > 0$. By (5),

$$f(a + nd) = a + (n + 1)d \quad (n \geq 0). \quad (6)$$

Claim. For every $x \geq a + d$,

$$d - \frac{d^2}{4x} \leq g(x) \leq d + \frac{d^2}{16x - 4d}. \quad (7)$$

Proof. The set $\{a + (n+1)d : n \geq 0\} = \{a+d, a+2d, \dots\}$ is an arithmetic progression with step d whose least element is $a+d \leq x$. Let $m = \lfloor (x-a)/d \rfloor$; then $m \geq 1$ and $a+md \leq x < a+(m+1)d$, and x lies within $d/2$ of one of the two endpoints $a+md$, $a+(m+1)d$, both of which belong to the set. Hence there is an index $n \geq 0$ with

$$|a + (n+1)d - x| \leq \frac{d}{2}.$$

Write $y := a + nd$ and $\varepsilon := (y+d) - x$, so $f(y) = y+d = x+\varepsilon$ by (6) and $|\varepsilon| \leq d/2$. Note $x \geq a+d > d$, hence $|\varepsilon|/x < \frac{1}{2}$.

Lower bound. By (2), $(f(x) + y)^2 \geq 4xf(y) = 4x(x+\varepsilon)$; since $f(x) + y > 0$,

$$f(x) \geq 2\sqrt{x(x+\varepsilon)} - y = 2x\sqrt{1 + \frac{\varepsilon}{x}} - (x + \varepsilon - d).$$

By (E1) with $u = \varepsilon/x \in (-\frac{1}{2}, \frac{1}{2})$,

$$2x\sqrt{1 + \frac{\varepsilon}{x}} \geq 2x\left(1 + \frac{\varepsilon}{2x} - \frac{\varepsilon^2}{2x^2}\right) = 2x + \varepsilon - \frac{\varepsilon^2}{x},$$

so

$$f(x) \geq 2x + \varepsilon - \frac{\varepsilon^2}{x} - x - \varepsilon + d = x + d - \frac{\varepsilon^2}{x} \geq x + d - \frac{d^2}{4x},$$

using $\varepsilon^2 \leq d^2/4$. This is the left inequality of (7).

Upper bound. By (1), $(f(x) + y)^2 \leq 2x^2 + 2f(y)^2 = 2x^2 + 2(x+\varepsilon)^2$. A direct expansion gives the identity

$$2x^2 + 2(x+\varepsilon)^2 = (2x+\varepsilon)^2 + \varepsilon^2.$$

Since $2x+\varepsilon \geq 2x - \frac{d}{2} > 0$, (E2) with $A = 2x+\varepsilon$, $B = \varepsilon^2$ yields

$$f(x) \leq \sqrt{(2x+\varepsilon)^2 + \varepsilon^2} - y \leq 2x + \varepsilon + \frac{\varepsilon^2}{2(2x+\varepsilon)} - (x + \varepsilon - d) = x + d + \frac{\varepsilon^2}{2(2x+\varepsilon)}.$$

Finally $\varepsilon^2 \leq d^2/4$ and $2(2x+\varepsilon) \geq 2(2x - \frac{d}{2}) = 4x - d > 0$, so

$$f(x) \leq x + d + \frac{d^2/4}{4x-d} = x + d + \frac{d^2}{16x-4d},$$

the right inequality of (7). $\square$

Step 4: all positive gaps are equal. Suppose $g(a) = d > 0$ and $g(a') = d' > 0$. For every $x \geq \max(a+d, a'+d')$, (7) applied to both points gives

$$d - \frac{d^2}{4x} \leq g(x) \leq d' + \frac{d'^2}{16x-4d'},$$

so that $d - d' \leq \frac{d^2}{4x} + \frac{d'^2}{16x-4d'}$ for all such x . The right-hand side tends to 0 as $x \rightarrow \infty$, whence $d \leq d'$; by symmetry $d' \leq d$. Hence $d = d'$:

$$(8) \quad \text{all positive values of } g \text{ are equal.}$$

Step 5: if some gap is positive, then $g \equiv d$ on a half-line. Assume $g(a) = d > 0$. For every $x \geq a + d$ we have $x > d > \frac{d}{4}$, so the lower bound in (7) gives

$$g(x) \geq d - \frac{d^2}{4x} > 0.$$

By (8), a positive gap must equal d ; hence

$$f(x) = x + d \quad \text{for all } x \geq T := a + d. \quad (9)$$

Step 6: downward bootstrap — $f(x) = x + d$ on all of $\mathbb{R}_{>0}$. **Claim.** If $f(t) = t + d$ for all $t \geq T$ (some $T > 0$, $d > 0$), then $f(y) = y + d$ for every $y > 0$ with $y \geq T - d$.

Proof. Let $y > 0$ with $y + d \geq T$, and put $x := y + d \geq T$, so $f(x) = x + d = y + 2d$. - Inequality (1) with this pair: $2(y+d)^2 + 2f(y)^2 \geq (y+2d+y)^2 = 4(y+d)^2$, hence $f(y)^2 \geq (y+d)^2$, i.e. $f(y) \geq y + d$. - Inequality (2) with this pair: $(y + 2d + y)^2 \geq 4(y + d)f(y)$, i.e. $4(y + d)^2 \geq 4(y + d)f(y)$, hence $f(y) \leq y + d$.

Therefore $f(y) = y + d$. $\square$

Starting from (9) and applying the Claim repeatedly, we get by induction on $k \geq 0$: $f(y) = y + d$ for all $y > 0$ with $y \geq T - kd$. Indeed, the base case $k = 0$ is (9); for the inductive step, if $f(t) = t + d$ holds for all $t > 0$ with $t \geq T - kd$, then for any $y > 0$ with $y \geq T - (k + 1)d$ the point $x = y + d$ satisfies $x > 0$ and $x \geq T - kd$, so $f(x) = x + d$ and the two displayed computations above give $f(y) = y + d$. Since $T - kd \rightarrow -\infty$, every $y > 0$ is covered:

$$f(y) = y + d \quad \text{for all } y > 0.$$

Step 7: conclusion. Either $g(y) = f(y) - y$ is identically 0, in which case $f(y) = y = y + 0$; or some gap is positive, in which case Steps 3–6 give $f(y) = y + d$ for all $y > 0$ with the constant $d > 0$. In both cases

$$f(x) = x + c \quad \text{for all } x > 0, \text{ for some constant } c \geq 0.$$

Together with Part I, the set of solutions is exactly $\{x \mapsto x + c : c \geq 0\}$. $\blacksquare$